Exploratory analysis of the 2021 BRFSS Survey

Project report for *Data Science for Health Systems*

1st Nicola Cucina
Dipartimento di Ingegneria
Università degli Studi di Perugia
Perugia, Italia
nicola.cucina@studenti.unipg.it

2nd Aurora Traversini
Dipartimento di Ingegneria
Università degli Studi di Perugia
Perugia, Italia
aurora.traversini@studenti.unipg.it

***Abstract*—Each year, the BRFSS collects data about U.S. residents' health. This is done via a telephone survey where various questions spanning 15 different core sections are asked. This report analyses the data collected in 2021 to investigate how different factors relate to a diabetes self-reported diagnosis. To perform the analysis, out of 303 features only 25 were selected, as they were considered the most significant for an initial screening for diabetes diagnosis.**

Through the evaluation of three groups of subjects, i.e. those with Diabetes, without Diabetes and with Prediabetes, what emerges is that the groups present a substantially different behaviour one from the other, though it isn't clear if this depends on the *unsuitability* of the dataset to the standard pipeline used in statistical analysis or if the observed trends are real.

***Index Terms*—diabetes, EDA, CDC, BRFSS**

I. Introduction

Diabetes is a chronic medical condition that arises when the pancreas fails to produce sufficient insulin or when the body cannot effectively utilize the insulin it produces. Insulin is crucial for regulating blood sugar levels [1]. Uncontrolled diabetes often results in hyperglycemia, a condition characterized by elevated blood glucose levels, which over time can cause severe damage to various body systems, particularly the nerves and blood vessels.

In 2014, 8.5% of adults aged 18 and above were diagnosed with diabetes. By 2019, diabetes was responsible for 1.5 million deaths, with 48% of these fatalities occurring before the age of 70 [2]. Additionally, diabetes contributed to 460,000 deaths from kidney disease and was a factor in approximately 20% of cardiovascular deaths [3].

From 2000 to 2019, the age-standardized mortality rate due to diabetes saw a 3% increase. In lower-middle-income countries, this mortality rate surged by 13%. Conversely, the global probability of dying from one of the four major noncommunicable diseases (cardiovascular diseases, cancer, chronic respiratory diseases, or diabetes) between the ages of 30 and 70 decreased by 22% during the same period. In the following study the data collected in 2021 will be analysed to investigate how different factors relate to a diabetes diagnosis among the various subjects who responded to the survey questions.

II. Dataset

A. Dataset Description

The data used in this study is obtained from the web portal of the Behavioral Risk Factor Surveillance System (BRFSS), USA's "premier system of health-related telephone surveys that collect state data about U.S. residents regarding their health-related risk behaviours, chronic health conditions, and use of preventive services" [4]. Each year, the BRFSS conducts over 400,000 interviews with adults, establishing it as the largest ongoing health survey system globally [4]. The dataset, entitled '**2021 BRFSS Survey Data**', is provided in a format suitable for use with SAS (Statistical Analysis System), software developed by SAS Institute, Inc..

The original dataset comprises 303 features, each representing a question or a calculated variable from the survey [5]. The questions are categorized into the following 15 thematic sections: *Health Status*, *Healthy Days*, *Health Care Access*, *Exercise*, *Hypertension Awareness*, *Cholesterol Awareness*, *Chronic Health Conditions*, *Arthritis*, *Demographics*, *Disability*, *Tobacco Use*, *Alcohol Consumption*, *Immunization*, *H.I.V./AIDS*, and *Fruits and Vegetables*.

The BRFSS survey uses a weighting methodology to ensure that the samples accurately represent the population and it includes several calculated variables [6]. The creation of these variables involves varying levels of complexity, from simply merging some of the values to more intricate combinations where multiple variables are confronted simultaneously [6]. Moreover, most variables derived from the BRFSS interviews have a *refused* code category, typically represented by the values 9, 99, or 999. These values can also indicate missing responses, which may result from non-interviews, i.e. where an interview ends before the question is asked, or from skip patterns in the questionnaire. Additionally, these codes may capture instances where an answer was expected but not provided, appearing as blank or another symbol [6].

B. Dataset Modeling

Before moving to the Data Analysis and Statistical Tests, the dataset has been reduced by selecting the following features:

- **DIABETE4**, renamed as **Diabetes**, is the label used in the dataset to differentiate between people *with Diabetes* (1), *without Diabetes* (2) and *with Prediabetes* (3).

- **X_RFHYPE6**, renamed as **HighBloodPressure**, represents people that have either high blood pressure (1) or not (2).
- **TOLDHI3**, renamed as **ToldHighColesterol**, represents whether people were told by a doctor that they have high cholesterol levels (1) or not (2).
- **CHOLCHK3**, renamed as **CholesterolCheck**, indicates how often the cholesterol levels are checked, divided into: *Less than a year* (1), *Less than two years* (2), *Less than three years* (3), *Less than four years* (4), *Less than five years* (5), *Five years or more* (6) or *Never* (7).
- **X_BMI5CAT**, renamed as **BMI**, represents the weight category of a person: *Underweight*(1), *Normal weight* (2), *Overweight* (3) or *Obese* (4).
- **X_SMOKER3**, renamed as **Smoker**, classifies each person based on how often they smoke: *Never* (1), *Former Smoker* (2), *Someday Smoker* (3) or *Everyday Smoker* (4).
- **X_MICHD**, renamed as **HeartDisease**, indicates whether people have ever reported having coronary heart disease or myocardial infarction (1) or they did not report having such conditions (2).
- **CVDSTRK3**, renamed as **Stroke**, represents whether people were told that they have ever had a stroke (1) or not (2).
- **X_TOTINDA**, renamed as **PhysicalActivity**, represents whether adults reported doing physical activity or exercise during the past month (1) or not (2).
- **X_FRTLT1A**, renamed as **Fruit**, represents whether adults have reported eating at least one fruit per day (1) or haven't (2).
- **X_VEGLT1A**, renamed as **Vegetables**, represents wheter adults have reported eating vegetables at least once per day (1) or haven't (2).
- **X_RFDRHV7**, renamed as **HeavyDrinker**, represents whether adults had more than 14 drinks per week (men) or more than 7 drinks per week (women) (1) or haven't (2).
- **X_HLTHPLN**, renamed as **HealthPlan**, represents whether adults have some form of health insurance (1) or none (2).
- **MEDCOST1**, renamed as **MedicalCost**, asks if there was a time in the past 12 months when a visit to a doctor was necessary but not affordable (1) or if there was no such time (2).
- **GENHLTH**, renamed as **GeneralHealth**, questions people if they think their health, in general, is: *Excellent* (1), *Very Good* (2), *Good* (3), *Fair* (4) or *Poor* (5).
- **MENTHLTH**, renamed as **MentalHealth**, asks for how many days during the past month their mental health was not good.
- **PHYSHLTH**, renamed as **PhysicalHealth**, asks for how many days during the past month your physical health was not good.
- **DIFFWALK**, renamed as **DifficultyWalking**, asks partecipants whether they have serious difficulty walking or climbing stairs (1) or not (2).
- **SEXVAR**, renamed as **Sex**, represents the sex of the respondent: *Male* (1) or *Female* (2).
- **X_AGEG5YR**, renamed as **Age**, places each individual in a 5-year age interval: *[18 - 24]* (1), *[25 - 29]* (2), *[30 - 34]* (3), *[35 - 39]* (4), *[40 - 44]* (5), *[45 - 49]* (6), *[50 - 54]* (7), *[55 - 59]* (8), *[60 - 64]* (9), *[65 - 69]* (10), *[70 - 74]* (11), *[75 - 79]* (12) or *80+* (13).
- **X_EDUCAG**, renamed as **Education**, represents the highest grade of education achieved: *College degree* (1), *Attended college* (2), *High school graduate* (3) or *Attended high school* (4).
- **X_INCOMG1**, renamed as **Income**, asks what the annual household income from all sources is, placed in the following intervals: *200k+* (1), *between 100k and 200k* (2), *between 50k and 100k* (3), *between 35k and 50k* (4), *between 25k and 35k* (5), *between 15k and 25k* (6) or *Less than 15k* (7).
- **CHECKUP1**, renamed as **Checkup**, represents how long ago the last routine checkup was: *Within past year* (1), *Within past 2 years* (2), *Within past 5 years* (3), *5 or more years ago* (4) or *Never* (5).
- **BLDSUGAR**, renamed as **BloodSugar**, indicates how often the participant's blood is checked for glucose or sugar in a year.
- **FEETCHK3**, renamed as **FeetCheck**, indicates how often the participant's feet are checked for sores or irritations in a year.

Features whose name begins with **X_** are *calculated variables*, and inside the codebook is indicated the logic behind each one (in the codebook these variables are indicated as **_NAME**, without the X at the beginning) [7].

This specific subset of features was chosen by going over all the columns in the dataset and looking for the most significant and relevant ones in various *Core Sections* defined in the questionnaire. At this stage, it was not checked whether all the questions were actually asked to each interviewee and therefore if the dataset was complete or not. The interview follows a structure defined in the **2021 BRFSS Questionnaire** and it may happen that, depending on the various answers obtained or on the *Optional Modules* used, some of the questions may be skipped or may be left out, leading to *BLANK* values in the dataset. An example of this issue are the features **BloodSugar** and **FeetCheck** which belong to the *Optional Module n.1 - Prediabetes* [8]. The chosen features are ultimately obtained from the original dataset by collapsing some of the classes to simplify the values used. When possible, the value *0* was chosen to represent *Don't know / Not sure / Refused / Blank*, but for the features **MentalHealth** and **PhysicalHealth** the value *31* is used since *30* is the maximum reported amount, while for **BloodSugar** and **FeetCheck** the value *42069* is used since it is well above any expected result.

III. Analysis

In this section, various relevant features of the dataset will be analysed, starting with the **Diabetes** column to gain a

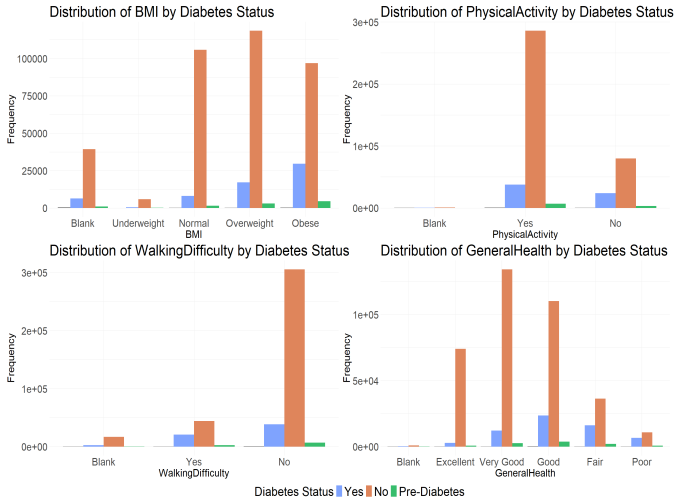


Figure 1: **BMI**, **GeneralHealth**, **PhysicalActivity** and **WalkingDifficulty** distribution

general understanding of its distribution.

Table I: Diabetes Distribution

	Null	With Diabetes	No Diabetes	PreDiabetes
Count	985	61424	366342	9942
Perc.	0.22453%	14.00159%	83.50760%	2.26627%

As shown in Table I the dataset is heavily disproportionate towards the **No Diabetes** group, which will lead to problems in the analysis in further sections.

Among the various features available, Figure 1 shows the distributions of some of the most interesting ones, as they present more significant variability with respect to the **Diabetes** variable.

From the plots in Figure 1 it is evident that individuals classified as *Obese* or *Overweight* appear to have a stronger link with the Diabetes condition compared to the those in the other classes. Similarly, individuals who do not engage in Physical Activity tend to present a higher probability of having Diabetes. Additionally, it can be seen that individuals without Diabetes predominantly do not experience difficulty in walking.

In Figure 2 two pairs of variables that are logically related to each other (i.e., **HighBloodPressure** with **ToldHighColesterol** and **HeartDisease** with **Stroke**) are displayed, in relation to the presence or absence of Diabetes.

The plots in Figure 2 suggest that individuals *with Diabetes* tend to have higher impact on the **ToldHighColesterol** and **HighBloodPressure** columns compared to those *without Diabetes*. Similarly, it indicates that individuals *without Diabetes* are more likely to have not experienced a stroke or some form of heart disease.

The observed differences in the distributions of **Diabetes** do not automatically indicate statistical differences among the

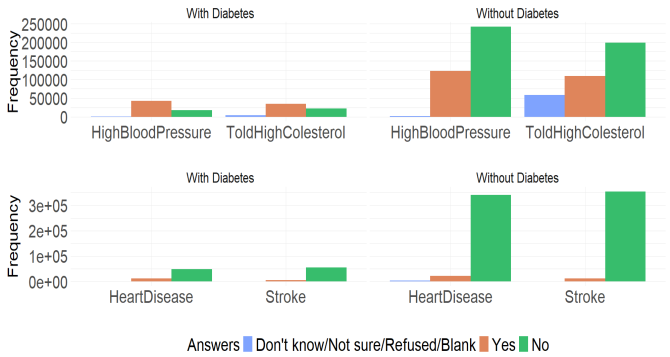


Figure 2: **HighBloodPressure-ToldHighColesterol** and **HeartDisease-Stroke** distribution grouped by the **Diabetes** groups

groups. To rigorously examine these differences, a statistical analysis will be conducted, dividing the variables into three groups: *Categorical*, *Numerical*, and *Ordinal*. This approach ensures that each type of variable is analysed accordingly to its specific characteristics. For this reason, three distinct R scripts were used, in order to have each script focused on a specific group of variable.

A. Categorical variables

As described in the dataset modeling section II-B, the categorical features in the dataset are: **Diabetes**, **HighBloodPressure**, **ToldHighColesterol**, **HeartDisease**, **Stroke**, **PhysicalActivity**, **Fruit**, **Vegetables**, **HeavyDrinker**, **HealthPlan**, **MedicalCost**, **WalkingDifficulty** and **Sex**.

Since the variables considered in this section are qualitative, the only statistical tests that can be used are the χ^2 test and Fisher's exact test.

1) Spearman's rank correlation: It was decided to begin with the Spearman's rank correlation to examine the relationships between pairs of categorical variables since it is the appropriate form of correlation for this type of features.

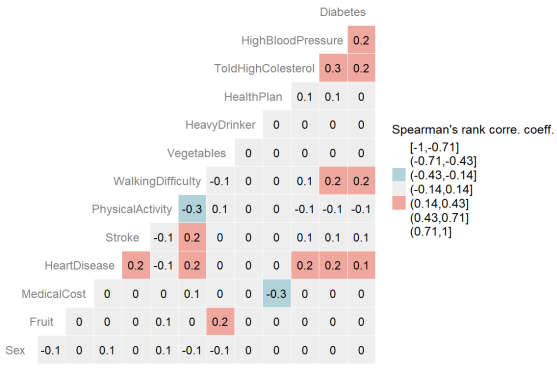


Figure 3: Spearman rank correlation matrix

As shown in Figure 3, the correlation between pairs of categorical variables is often *null* or very low, suggesting

that there is weak or no relationship between most of the categorical variables in the dataset.

2) χ^2 test of independence and Fisher test: To check whether the two considered features are indipendent or not, the χ^2 and the *Fisher* tests were used, both having that the *null hypothesis* is that the two variables are independent.

To perform these two tests, all pairs of the 13 categorical variables were considered, resulting in 78 combinations.

Table II: χ^2 and Fisher test results

Feature1	Feature2	χ^2p-value	Fisher p-value
HeartDisease	Fruit	0.08319371	0.08232534
Stroke	Fruit	0.17990670	0.17779149
Stroke	Sex	0.14011637	0.13853025

p -values obtained from the Chi-squared test and the Fisher exact test are very similar but are reported for completeness

Only the p -values exceeding the significance threshold of 5% are shown in Table II, indicating that the null hypothesis of independence can be rejected for all the pairs of variables except the ones reported.

To understand the distribution of the reported variables, bar charts were used, combining each feature with the **Diabetes** label to better comprehend the differences between the groups.

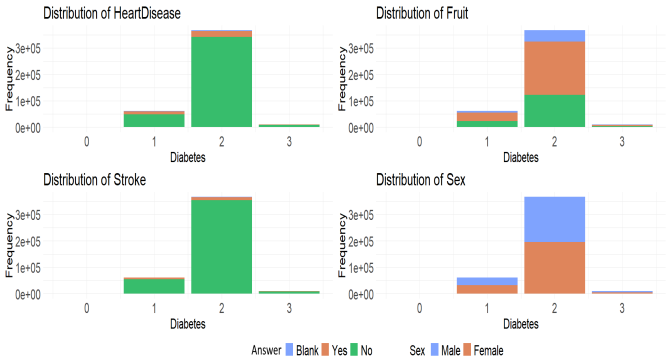


Figure 4: **HeartDisease**, **Fruit**, **Stroke** and **Sex** Distribution

The results obtained via the χ^2 and Fisher tests are not observable from the various graphs seen in figure 4, since only **HeartDisease** and **Stroke** seem to behave in a similar fashion.

B. Ordinal variables

As described in the dataset modeling section II-B, the ordinal features in the dataset are: **CholesterolCheck**, **BMI**, **Smoker**, **GeneralHealth**, **Age**, **Education**, **Income** and **Checkup**.

Given that more than two groups are being compared (*with Diabetes*, *without Diabetes*, *Prediabetes*) and the measurements are *independent* (i.e., no repeated measures), the appropriate statistical tests are the *Kruskal-Wallis* test for assessing differences between groups, followed by *Dunn*'s post-hoc test with Bonferroni correction for pairwise comparisons.

1) Spearman's rank correlation: Before conducting these tests, the relationships between pairs of ordinal variables will be examined using Spearman's rank correlation, as seen before in section III-A1.

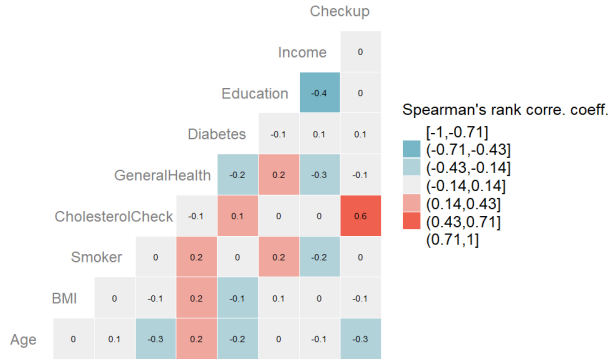


Figure 5: Spearman rank correlation matrix

As one can see in Fig. 5, the correlations obtained are generally low, indicating weak relationships between most pairs of variables, with just a few of them more significant than others.

To better understand these results, it is important to note that the ordinal data has been ordered in section II-B from the *least severe* to the *most severe* level in terms of the associated level of risk.

2) Kruskal-Wallis Test: The Kruskal-Wallis test was performed to determine if there are statistically significant differences between the three groups, since the *null hypothesis* for this test is that the *median* of each group is the same.

Table III: Kruskal-Wallis test results

Feature	χ^2	p-value
CholesterolCheck	13837	$<$ 2.2e-16
BMI	14777	$<$ 2.2e-16
Smoker	938.72	$<$ 2.2e-16
GeneralHealth	30673	$<$ 2.2e-16
Age	17627	$<$ 2.2e-16
Education	3473	$<$ 2.2e-16
Income	6651	$<$ 2.2e-16
Checkup	9057	$<$ 2.2e-16

As shown in Table III, the null hypothesis could be rejected in all cases as the p-values are extremely low. This indicates significant differences among the groups. The consistently low p-value of $<$ 2.2e-16 is a limit of precision in R that indicates that the result of a computation may no longer be reliable and it acts as a default threshold to indicate an extremely small p-value, beyond which the exact value becomes less relevant [9].

3) Dunn post hoc Test with Bonferroni correction: After the results of the Kruskal-Wallis test showed in section III-B2, Dunn's post-hoc test was performed, since this test aims to identify specific pairwise differences between the groups.

Table IV: Dunn test results

Column	Groups	Z	Unadj. p-value	Adj. p-value
Education	1 - 3	2.283837	2.2381e-02	6.7143e-02
Smoker	1 - 3	2.968583	2.9917e-03	8.9752e-03
Income	1 - 3	-3.081331	2.0607e-03	6.1823e-03

Out of all the 24 comparisons computed (*3 groups compared across 8 columns*), only the p-values that are "close enough" to the threshold are reported in table IV. Even then, only the **Education** column has a p-value bigger than the required 5% which allows the null hypothesis to be accepted, i.e. the groups 1 (*with diabetes*) and 3 (*with prediabetes*) are similar. In all the other cases, the result obtained by the Kruskal-Wallis test are confirmed.

C. Numerical variables

As for the numerical variables in the dataset, as explained in section II-B, these are: **MentalHealth**, **PhysicalHealth**, **BloodSugar**, **FeetCheck**. This kind of features requires a different analysis since the *normality* property, if present, leads to completely different pipeline of statistical tests that need to be performed.

1) Pearson's correlation: Given that these features are numerical, rather than using Spearman's rank correlation as done previously, computing Pearson's correlation coefficient is more appropriate for assessing the linear relationships between each feature and the Diabetes condition.

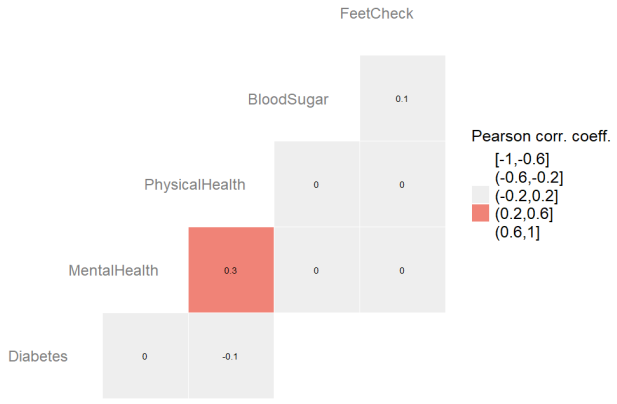


Figure 6: Pearson correlation matrix

The matrix in Figure 6 shows weak, if any, correlation between the variable **Diabetes** with both **MentalHealth** and **PhysicalHealth**. However, it can be noted that the correlation analysis performed for **Diabetes** with the variables **BloodSugar** and **FeetCheck** resulted in *NA* values.

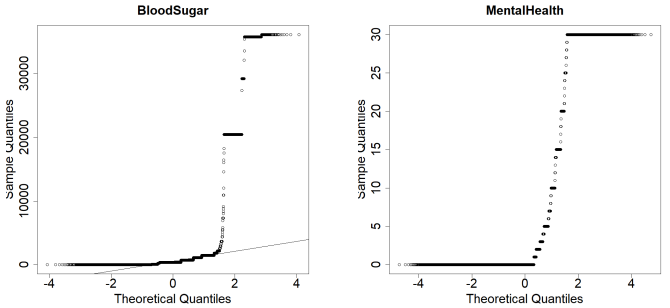
Table V: Group distribution

Diabetes	BS	FC	BS missing	FC missing
null	0	0	985	985
1	21523	21250	39901	40174
2	0	0	366342	366342
3	0	0	9942	9942

BS = **BloodSugar**, FC = **FeetCheck**, both filtered to exclude missing values and include only the valid ones (42069 being the threshold)

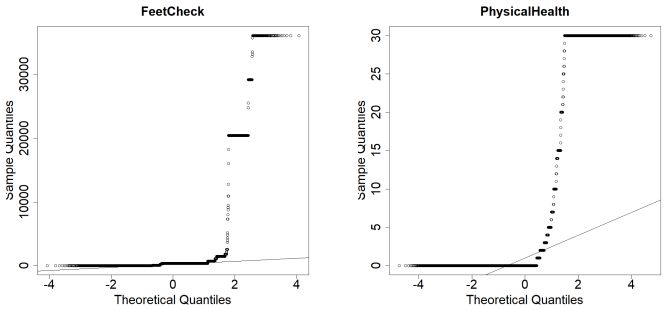
As shown in table V, these values only appear in the presence of the group 1, i.e., *with Diabetes*, and not for the other two groups, preventing their correlation from being evaluated. This is a problem that stems from the questionnarie structure and how the interviews are conducted that cannot be resolved in this study.

2) Normality check: To assess the normality of the numerical features, Q-Q plots will first be shown to get a general idea of their distributions. After this visual inspection, formal statistical tests will be performed to assess the normality of the data more rigorously.



(a) Q-Q Plot for BloodSugar

(b) Q-Q Plot for MentalHealth



(c) Q-Q Plot for FeetCheck

(d) Q-Q Plot for PhysicalHealth

Figure 7: Q-Q Plots for assessing normality.

As shown in Figure 7, the variables do not follow a normal distribution. Since the Shapiro-Wilk test is not applicable due to the large sample size, the Anderson-Darling (AD) test is employed to evaluate the normality of the data.

Table VI: Anderson-Darling test results

Diabetes	MentalHealth	PhysicalHealth
null	**A**=158.31, **p-value** $<$ 2.2e-16	**A**=158.37, **p-value** $<$ 2.2e-16
1	**A**=11080, **p-value** $<$ 2.2e-16	**A**=9091.3, **p-value** $<$ 2.2e-16
2	**A**=66069, **p-value** $<$ 2.2e-16	**A**=81205, **p-value** $<$ 2.2e-16
3	**A**=1711.3, **p-value** $<$ 2.2e-16	**A**=1733.9, **p-value** $<$ 2.2e-16

A = *Anderson-Darling* statistic computed using the column **MentalHealth** or **PhysicalHealth**, divided by the groups *with Diabetes* (1), *without Diabetes* (2) or *with Prediabetes* (3)

The p-values obtained in table VI confirm that the data do not follow the normal distribution. The Anderson-Darling test is not applicable to the **BloodSugar** and **FeetCheck** features given the distribution shown in table V.

3) Kruskal-Wallis Test: a Kruskal-Wallis test is necessary to continue the analysis. As explained before in section III-B2, this test is performed to determine if there are statistically significant differences between the various groups.

Table VII: Kruskal-Wallis test results

Feature	χ^2	p-value
MentalHealth	63.833	8.913e-14
PhysicalHealth	8713.7	$<$ 2.2e-16
BloodSugar	*NA*	*NA*
FeetCheck	*NA*	*NA*

Using the logic explained before in the table V, the **BloodSugar** and **FeetCheck** features cannot be tested this way, while the p-value found for **MentalHealth** and **PhysicalHealth** leads us to refuse the null hypothesis.

4) Dunn post hoc Test: To check wheter the results found in section III-C3 are significant, the Dunn post hoc test is performed.

Table VIII: Dunn test results

Column	Group	Z	Unadj. p-value	Adj. p-value
MH	1 - 2	4.981980	6.293689e-07	1.888107e-06
MH	1 - 3	-4.058353	4.9420e-05	1.4826e-04
MH	2 - 3	-6.442975	1.1715e-10	3.5146e-10
PH	1 - 2	91.39737	0.0000e+00	0.0000e+00
PH	1 - 3	14.35440	9.9990e-47	2.9997e-46
PH	2 - 3	-23.88365	4.5292e-126	1.3587e-125

MH = **MentalHealth**, PH = **PhysicalHealth**

As one can see in table VIII, the **BloodSugar** and **FeetCheck** features are missing since the Kruskal-Wallis test failed, while for the features **MentalHealth** and **PhysicalHealth** the results found in section III-C3 are confirmed.

IV. Conclusions

In conclusion, the results obtained from the statistical analysis of the dataset suggest a significant difference between the distributions. However, it is unclear whether these differences are due to actual differences within the population or if they stem from limitations in the dataset itself.

It is important to note that the nature of the dataset, which was obtained through a telephone survey, hindered the application of certain statistical tools, thus limiting its utility for diabetes prevention efforts. The dataset's structure is more suited for providing a yearly snapshot of the population rather than supporting in-depth statistical analysis.

This study highlights that the dataset is not designed for a comprehensive statistical analysis aimed at prevention strategies for the following reasons:

- due to the structure of the questionnaire, there are a lot of rows with *null* values, as seen in Table V. This limits the number of effective rows of the dataset that can be used to compute the test results;
- the large sample size makes it more difficult for the data to follow the normal distribution. This represents a problem since many statistical tests need the normality property to give significant results;
- it is known that feature selection introduces a bias and even if our hypothesis is that this does not represent a valid reason for the problems encountered, it has not been tested whether these problems persist even when using the whole dataset.

For more robust and statistically significant results, it would be beneficial to combine and analyse the datasets collected by the BRFSS over multiple years. This approach would allow for observing trends and changes in the population and its characteristics over a longer period, thereby providing more valuable insights for diabetes prevention.

References

[1] World Health Organization. (2024). "Diabetes". Recuperato da https://www.who.int/news-room/fact-sheets/detail/diabetes

[2] Samar A. Antar and Nada A. Ashour and Marwa Sharaky and Muhammad Khattab and Naira A. Ashour and Roaa T. Zaid and Eun Joo Roh and Ahmed Elkamhawy and Ahmed A. Al-Karmalawy, Diabetes mellitus: Classification, mediators, and complications; " A gate to identify potential targets for the development of new effective treatments ", Biomedicine & Pharmacotherapy, vol.168, pp.115734, 2023, doi: 10.1016/2023/115734.

[3] Global Burden of Disease Collaborative Network, "Global Burden of Disease Study 2019: Results," Institute for Health Metrics and Evaluation, 2020. [Online]. Available: https://vizhub.healthdata.org/gbd-results/

[4] Behavioral Risk Factor Surveillance System. (2024). "Behavioral Risk Factor Surveillance System". Retrieved from https://www.cdc.gov/brfss/

[5] CDC-BRFSS, July 2022, "2021 BRFSS Survey Data", CDC-BRFSS, [Online]. Available: https://www.cdc.gov/brfss/annual_data/annual_2021.html.

[6] CDC-BRFSS, *Overview: BRFSS 2021*, pp.7-9, (July 2022), Accessed 2024. [Online]. Available: https://www.cdc.gov/brfss/annual_data/2021/pdf/Overview_2021-508.pdf.

[7] CDC-BRFSS, *LLCP 2021 Codebook Report*, (August 2022), Accessed 2024. [Online]. Available: https://www.cdc.gov/brfss/annual_data/2021/pdf/codebook21_llcp-v2-508.pdf

[8] CDC-BRFSS, *2021 BRFSS Questionnaire*, (August 2022), Accessed 2024. [Online]. Available: https://www.cdc.gov/brfss/questionnaires/pdf-ques/2021-BRFSS-Questionnaire-1-19-2022-508.pdf

[9] Glen_b, *How should tiny p-values be reported? (and why does R put a minimum on 2.22e-16?)*, (April 2024), Accessed 2024. [Online]. Available: https://stats.stackexchange.com/q/78840